

Study Design Overview

COHORT

Extreme Phenotype Cohort (EPC): $n = 52$

High-grade, localized, treatment-naïve PCa with ≥ 7 years follow-up

Whole-exome sequencing (tumor + matched benign tissue) gnsRVs: $\text{MAF} \leq 2\%$

Metastatic
($n = 26$)

Matched: ancestry, GS, PSA,
stage, margins, follow-up

Non-metastatic
($n = 26$)

DISCOVERY

Variant discovery

PRESPECIFIED

PRIMARY HYPOTHESIS TEST (prespecified)
25 canonical DDR genes
Bootstrapping + Wilcoxon rank-sum

EXPLORATORY

SECONDARY EXPLORATORY SCREEN (genome-wide)
gnsRVs exclusive to ≥ 3 metastatic patients
CNV + survival analysis in public cohorts

81 met-exclusive gnsRVs in DDR genes ($p = 4.57 \times 10^{-6}$)
Synonymous variants: no difference (internal control)

Sensitivity analyses
Firth LR(adj. age+ancestry):
OR=25.0, $p < 0.001$
Follow-up MWU: $p < 0.001$
Burden $p = -0.12$, $p = 0.55$

56 gnsRVs in 53 genes ($p = 2.98 \times 10^{-8}$)
Somatic selection & survival signal in public PCa cohorts

REPLICATION

Independent replication

REPLICATION 1

Australian EPC ($n = 53$)
Carrier frequency > expected (binomial)
 $p = 0.03-0.05$ (metastatic) vs. $p = 0.45$ (non-metastatic)

REPLICATION 2

PPCG cohort ($n = 976$, including 200 metastatic)
Aggregate gnsRV score: high-risk disease (SKAT $p < 0.03$)
Gene-level: CHEK2, FOCAD, SPATA9, ZSWIM4, DNAJC10

3-criterion selection: exclusive to metastatic arm · PCa biological relevance · cell model availability

FUNCTIONAL

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Functional characterization (2 discovery-stage candidates selected prior to and independent of replication)
KDM6B K973Q [genome-wide screen, ≥ 3 metastatic carriers] → migration & invasion in LNCaP
BRCA2 I1962T [prespecified DDR candidate; ultrarare] → Olaparib sensitivity in HEK293T